

Revision of Linear Algebra

Advanced Statistical Inference

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Notation

- Vectors $\mathbf{v} \in \mathbb{R}^n$

$$\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$$

- Matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$

$$\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \cdots & A_{mn} \end{bmatrix}$$

- v_i indexes elements of $\mathbf{v}$ for some i
- A_{ij} indexes elements of $\mathbf{A}$ for some i and j

Basic matrix operations

- Matrix addition: $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$
- Scalar multiplication: $(\gamma \mathbf{B})_{ij} = \gamma b_{ij}$
- Matrix-vector multiplication:

$$(\mathbf{A}\mathbf{v})_i = \sum_{j=1}^n a_{ij}v_j$$

- Matrix-matrix multiplication:

$$(\mathbf{AB})_{ij} = \sum_{k=1}^n a_{ik}b_{kj}$$

💡 Tip

In Numpy, use `@` for both matrix-vector and matrix-matrix multiplication (depending on the dimensions of the arrays).

```
import numpy as np
A = np.array([[1, 2], [3, 4]])
B = np.array([[5, 6], [7, 8]])
v = np.array([1, 2])
```

```
C = A @ B
d = A @ v
```

```
print(f"C={C}")
print(f"d={d}")
```

```
C=[[19 22]
   [43 50]]
d=[ 5 11]
```

...

- Transpose $\mathbf{A}^\top$: $(\mathbf{A}^\top)_{ij} = a_{ji}$ and transpose of products $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$

Squared matrices

A square matrix is one where the number of rows equals the number of columns (e.g., $\mathbf{A} \in \mathbb{R}^{n \times n}$).

Properties:

- A squared matrix is symmetric if $\mathbf{A} = \mathbf{A}^\top$
- A squared matrix is orthogonal if $\mathbf{A}^\top \mathbf{A} = \mathbf{A} \mathbf{A}^\top = \mathbf{I}$
- A matrix is diagonal if $a_{ij} = 0$ for $i \neq j$

...

Trace

- Trace operator $\text{Tr}(\mathbf{A}) = \sum_{i=1}^n a_{ii}$
- Trace is invariant under cyclic permutations: $\text{Tr}(\mathbf{ABC}) = \text{Tr}(\mathbf{CAB}) = \text{Tr}(\mathbf{BCA})$

Matrix inverse

The inverse of a square matrix $\mathbf{A}$ is denoted $\mathbf{A}^{-1}$ and satisfies

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

Important

The inverse does not always exist. We will see later the conditions under which it does.

If $\mathbf{A}$ is invertible, then $\mathbf{A}$ is said to be **non-singular**. Otherwise, $\mathbf{A}$ is **singular**.

Additional properties:

- Inverse of product: $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- Inverse of diagonal matrix: $\mathbf{D}^{-1} = \text{diag}(\mathbf{d}^{-1})$, where $\mathbf{d} = (d_1, \dots, d_n)$ is the diagonal of $\mathbf{D}$
- Woodbury matrix identity: $(\mathbf{A} + \mathbf{UCV})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{VA}^{-1}\mathbf{U})^{-1}\mathbf{VA}^{-1}$

Determinant

The determinant of a square matrix $\mathbf{A}$ is denoted $|\mathbf{A}|$ or $\det(\mathbf{A})$ and is a scalar value.

- For a 2×2 matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is $ad - bc$.
- For a general $n \times n$ matrix, the determinant is defined recursively as

$$\det(\mathbf{A}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(\mathbf{A}_{-1,-j})$$

where $\mathbf{A}_{-1,-j}$ is the matrix $\mathbf{A}$ with the first row and j -th column removed. Note that $\mathbf{A}_{-1,-j}$ is a $(n-1) \times (n-1)$ matrix.

This is called the **cofactor expansion** of the determinant.

Intuition: The determinant can be interpreted as the signed volume of the parallelepiped spanned by the columns of $\mathbf{A}$.

Properties of determinants

- Given a triangular matrix $\mathbf{A} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ 0 & \ddots & \vdots \\ 0 & 0 & a_{nn} \end{bmatrix}$, then $\det(\mathbf{A}) = \prod_{i=1}^n a_{ii}$.
- Determinant of a sum of matrices: $\det(\mathbf{A} + \mathbf{B}) \neq \det(\mathbf{A}) + \det(\mathbf{B})$
- Determinant of a product: $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$
- Determinant of an inverse: $\det(\mathbf{A}^{-1}) = \det(\mathbf{A})^{-1}$
- Determinant of a transpose: $\det(\mathbf{A}^\top) = \det(\mathbf{A})$
- Determinant of a scalar multiple: $\det(\gamma\mathbf{A}) = \gamma^n \det(\mathbf{A})$
- A matrix is invertible if and only if its determinant is non-zero.
- Determinant of an orthogonal matrix: $\det(\mathbf{Q}) = \pm 1$

Spectral decomposition

Eigenvalues and eigenvectors

- Let $\mathbf{A}$ be an $n \times n$ matrix
- $\mathbf{A}$ is said to have an *eigenvalue* λ and (non-zero) *eigenvector* $\mathbf{v}$ corresponding to λ if

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$$

- Eigenvalues are the λ values that solve the determinantal equation $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$.
- Eigenvectors are the corresponding $\mathbf{v}$ values for which $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$.

Eigendecomposition

- The *Spectral theorem* says that every square and symmetric matrix $\mathbf{A}$ can be decomposed as

$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$$

- The columns of $\mathbf{Q}$ are the eigenvectors of $\mathbf{A}$
- The diagonal matrix $\mathbf{\Lambda}$ contains the eigenvalues of $\mathbf{A}$: $\mathbf{\Lambda} = \text{diag}(\boldsymbol{\lambda})$

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

- The eigenvectors may be chosen to be orthonormal, so that $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$.

Properties of eigendecomposition

We can use the eigendecomposition to compute the determinant of $\mathbf{A}$ as $\det(\mathbf{A}) = \prod_{i=1}^n \lambda_i$.

i Proof

By the spectral theorem, $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$, so $\det(\mathbf{A}) = \det(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top) = \det(\mathbf{Q})\det(\mathbf{\Lambda})\det(\mathbf{Q}^\top) = \det(\mathbf{\Lambda}) = \prod_{i=1}^n \lambda_i$.

...

The trace of $\mathbf{A}$ is the sum of its eigenvalues: $\text{Tr}(\mathbf{A}) = \sum_{i=1}^n \lambda_i$.

i Proof

By the spectral theorem, $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$, so $\text{Tr}(\mathbf{A}) = \text{Tr}(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top) = \text{Tr}(\mathbf{Q}^\top \mathbf{Q}\mathbf{\Lambda}) = \text{Tr}(\mathbf{\Lambda}) = \sum_{i=1}^n \lambda_i$.

Positive definite and semidefinite matrices

- The $n \times n$ matrix $\mathbf{A}$ is said to be **positive definite** if $\mathbf{y}^\top \mathbf{A} \mathbf{y} > 0$ for *all* $\mathbf{y} \neq \mathbf{0}$.
- The $n \times n$ matrix $\mathbf{A}$ is said to be **positive semidefinite** if $\mathbf{y}^\top \mathbf{A} \mathbf{y} \geq 0$ for *all* $\mathbf{y} \neq \mathbf{0}$.

A symmetric matrix is positive definite if and only if all its eigenvalues are positive.

Proof

Let $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ be the spectral decomposition of $\mathbf{A}$. Then $\mathbf{y}^\top \mathbf{A} \mathbf{y} = \mathbf{y}^\top \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top \mathbf{y} = (\mathbf{Q}^\top \mathbf{y})^\top \mathbf{\Lambda} (\mathbf{Q}^\top \mathbf{y}) = \sum_{i=1}^n \lambda_i (\mathbf{q}_i^\top \mathbf{y})^2$. Since $\lambda_i > 0$ for all i , then $\mathbf{y}^\top \mathbf{A} \mathbf{y} > 0$ for all $\mathbf{y} \neq \mathbf{0}$.

Corollary

If at least one eigenvalue of a symmetric matrix is zero, then the matrix is positive semidefinite.

Example: Show that $\mathbf{X}^\top \mathbf{X}$ is always positive semidefinite

Let $\mathbf{X}$ be an $n \times p$ matrix and $\mathbf{y}$ be a p -dimensional vector. Then

$$\begin{aligned} \mathbf{y}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{y} &= (\mathbf{X} \mathbf{y})^\top (\mathbf{X} \mathbf{y}) \\ &= \mathbf{z}^\top \mathbf{z} \\ &= \sum_{i=1}^n z_i^2 \geq 0 \end{aligned}$$

To guarantee that $\mathbf{X}^\top \mathbf{X}$ is positive definite, we also need to ensure that $\mathbf{X}$ is full rank.

In practice:

Sometimes, we **need** to ensure that $\mathbf{X}^\top \mathbf{X}$ is positive definite numerically. To ensure that we can add a small jitter to the diagonal, $\mathbf{X}^\top \mathbf{X} + \epsilon \mathbf{I}$, with $\epsilon > 0$.

Cholesky decomposition

- Cholesky decomposition is an algorithm to efficiently compute determinants and inverses of positive definite matrices.

Define lower triangular matrix $\mathbf{L}$

$$\mathbf{L} = \begin{bmatrix} L_{11} & 0 & \cdots & 0 \\ L_{21} & L_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ L_{n1} & L_{n2} & \cdots & L_{nn} \end{bmatrix}$$

so that $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$.

Question: How to go from $\mathbf{A}$ to $\mathbf{L}$?

Example of Cholesky algorithm: for 3×3 matrix $\mathbf{A}$:

$$\mathbf{L}\mathbf{L}^\top = \begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \begin{bmatrix} L_{11} & L_{21} & L_{31} \\ 0 & L_{22} & L_{32} \\ 0 & 0 & L_{33} \end{bmatrix} = \begin{bmatrix} L_{11}^2 & L_{21}L_{11} & L_{31}L_{11} \\ L_{21}L_{11} & L_{21}^2 + L_{22}^2 & L_{31}L_{21} + L_{32}L_{22} \\ L_{31}L_{11} & L_{31}L_{21} + L_{32}L_{22} & L_{31}^2 + L_{32}^2 + L_{33}^2 \end{bmatrix} \quad (\text{symmetric})$$

Properties of Cholesky decomposition:

- Iterative algorithm costing $\mathcal{O}(n^3)$ operations.
- The determinant of $\mathbf{A}$ is $\det(\mathbf{A}) = \det(\mathbf{L}\mathbf{L}^\top) = \det(\mathbf{L})^2 = (\prod_{i=1}^n L_{ii})^2$
- Usefull for solving $\mathbf{A}^{-1}\mathbf{b}$ (e.g., in linear systems and in linear regression) via back substitution.
- The inverse of $\mathbf{A}$ can be computed as back substitution

Cholesky decomposition for solving linear systems

To solve the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$, instead of computing $\mathbf{A}^{-1}\mathbf{b}$, we can use the Cholesky decomposition as follows:

1. Compute the Cholesky decomposition $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$
2. Solve the triangular system $\mathbf{L}\mathbf{y} = \mathbf{b}$ for $\mathbf{y}$ via forward substitution
3. Solve the triangular system $\mathbf{L}^\top\mathbf{x} = \mathbf{y}$ for $\mathbf{x}$ via back substitution

